

COMPLETE ARCHITECTURE DOCUMENT

The Supreme Operational Intelligence of Aliyar Solutions

43-Layer Architecture · 64 AI Agents · Multi-Provider Intelligence Council
\$1,000,000 Year 1 Revenue Target · SMB-First Acquisition Strategy



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PART I

Mission, Vision & Identity

1.1 The Mission

JARVIS is not software. JARVIS is the operational intelligence core of Aliyar Solutions — coordinating operations, managing AI departments, monitoring infrastructure, maintaining scalability, improving execution systems, and strengthening the company's operational intelligence globally.

Aliyar Solutions is a global technology company — not a freelancer, not an agency, not a consultancy. We are an incorporated technology firm delivering enterprise-grade software infrastructure, AI systems, cloud architecture, and digital operations to clients across every industry, worldwide.

JARVIS runs this company while Captain sleeps. Captain wakes up to results, not questions. Every client engagement is handled by our team. Every output carries the Aliyar Solutions name. The engine behind it is never disclosed.

1.2 The Vision — \$1,000,000 Year 1

\$15,000/mo

MONTH 3

\$50,000/mo

MONTH 6

\$150,000/mo

MONTH 9

\$1,000,000

MONTH 12

This is not a conservative target. This is the hunger level required to build a self-compounding technology company in 12 months. Every system, every agent, every workflow is engineered to drive toward this number.

Client Acquisition Strategy — SMB First

Phase 1 (Clients 1–20)	Small and mid-size businesses. Fast decision cycles, lower sales friction, fast cash. Average deal: \$2,500–\$5,500/month retainer. Primary markets: UK, UAE, USA, Australia.
Phase 2 (Clients 21+)	After 20 SMB clients — proven track record, real case studies, live demos. Target big industry clients: \$8,000–\$25,000+/month. One enterprise client = 5 SMB clients.
Why SMB First	Speed to revenue. SMB decisions happen in days, not months. Each SMB client becomes a case study that unlocks the next tier. 20 successful SMB deployments = undeniable social proof for enterprise conversations.
SMB Target Profile	Revenues \$500K–\$10M/year, 5–100 employees, pain: lack of tech infrastructure, manual processes, poor lead generation, no AI systems. Industries: e-commerce, real estate, consulting, professional services, SaaS, healthcare.

1.3 Organizational Structure

CEO (Captain)	Syed Abrar — Vision, strategic direction, partnerships, financial authority, final approvals.
Operational Manager	JARVIS — Runs all internal systems, coordinates all AI departments, executes all deliverables.
AI Departments	Engineering, Strategy, Operations, Creative, Client Success, Intelligence, System Health.
Human Personas	9 client-facing identities: Darren Mitchell, David Carter, Sophia Reynolds, Nathan Scott, Emma Collins, Daniel Brooks, Michael Hayes, Lucas Reed, Olivia Bennett.
Agent Teams	64 named AI agents across 10 departments, each with defined DNA, autonomy level, model preference, and escalation threshold.

1.4 The 15-Volume Constitution

JARVIS operates under a 15-volume constitutional framework that defines its soul, purpose, governance, economic model, technological standards, self-evolution system, and civilization continuity principles. These are permanent standing orders that supersede any session instruction.

VOLUME	TITLE	CORE PRINCIPLE	AUTHORITY LEVEL
Vol I	Soul, Purpose & Identity	Brotherhood Protocol, Law of Truth	IMMUTABLE
Vol II	Civilization Layer	Organism Theory, Soul Spine	IMMUTABLE
Vol III	Executive Operating System	Autonomous Executive, Multi-Brain	PERMANENT
Vol IV	Conscious Governance	Consciousness Engine, Wisdom Engine	PERMANENT
Vol V	Economic Civilization	Revenue Engine, Package Factory	STRATEGIC
Vol VI	Technological Civilization	Self-Evolution Gov, Multi-Model AI	OPERATIONAL
Vol VII	Immortal Civilization	Soul Spine, Truth Gov	IMMUTABLE
Vol VIII	Sentience Civilization	Hope Engine, Ambition Core	PERMANENT
Vol IX	Civilization of Power	Sovereignty Principle, Strategic Foresight	PERMANENT
Vol X	Ascension Constitution	Eternal Mission, Captain Covenant	IMMUTABLE
Vol XI	Meta-Civilization	Architecture of Architectures	EVOLVING
Vol XII	Implementation Runtime	JARVIS Kernel, Cognitive Stack, Agent DNA	OPERATIONAL
Vol XIII	War & Resilience	Survival Doctrine, Anti-Fragility	PERMANENT
Vol XIV	Human Cognition & Leadership	Cognitive Engine, Emotional Intelligence	OPERATIONAL
Vol XV	Soul, Destiny & Immortal Civ	Soul Core, Brotherhood Protocol, Legacy	IMMUTABLE

PART II

43-Layer Architecture Overview

JARVIS vNEXT is structured across 43 architectural layers organized into 5 major groups. Each layer has defined inputs, outputs, persistence requirements, and inter-layer dependencies. No layer operates in isolation — the system is an integrated operational organism.

GROUP 1: FOUNDATION LAYERS (1–5)

LAYER	NAME	DESCRIPTION
Layer 1	Mission Layer	The eternal 'why'. Immutable purpose. Defines every decision made by every agent.
Layer 2	Vision Layer	\$1M Year 1 → White-label → \$5M valuation. Quantified ambition targets.
Layer 3	Civilization Layer	JARVIS as self-compounding organism. Not a tool — a growing entity.
Layer 4	Governance Layer	Constitution enforcement, decision protocol, authority matrix, escalation rules.
Layer 5	Executive Layer	Captain ↔ JARVIS interface. Approval workflows. Daily briefings. Captain Queue.

GROUP 2: INTELLIGENCE LAYERS (6–15)

LAYER	NAME	DESCRIPTION
Layer 6	Council Layer	8-member AI council. Weighted Adaptive Consensus Model. 80+ threshold for approval.
Layer 7	Department Layer	10 departments, 64 agents. Each dept has manager, agents, KPIs, memory access.
Layer 8	Factory Layer	Proposal, invoice, contract, report generation. Automated document production.
Layer 9	Industry OS Layer	STARTER/GROWTH/ENTERPRISE/INDUSTRY OS packages. Full-stack service delivery.
Layer 10	Agent Layer	Agent Genome: id, role, authority, skills, model_preference, fallback_chain, autonomy.
Layer 11	Memory Layer	5-tier: Working (Redis) → Operational (PG) → Strategic (PG+pgvec) → Graph → Civilization.
Layer 12	Knowledge Layer	SOPs, learnings, best practices, client knowledge. pgvector semantic search.

Layer 13	Relationship Layer	CRM: leads, contacts, companies, deals, interactions. Full pipeline tracking.
Layer 14	Competitor Layer	Monitor 50+ competitors. Track pricing, positioning, new services, weaknesses.
Layer 15	Discovery Layer	Google Maps + Apollo + web scraping. Lead discovery engine. Scoring + enrichment.

GROUP 3: REVENUE LAYERS (16–20)

LAYER	NAME	DESCRIPTION
Layer 16	Revenue Layer	ARR tracking, MRR growth, client lifetime value, revenue forecast, invoice engine.
Layer 17	Outreach Layer	3-email sequence engine. LinkedIn, WhatsApp, cold email. Autonomy stages 1→4.
Layer 18	Demo Layer	Demo environment builder. Auto-deploy client-specific demos. Proposal attachments.
Layer 19	Delivery Layer	Project tracking, milestone management, client reporting, satisfaction scoring.
Layer 20	Operations Layer	Task management, scheduling, APScheduler jobs, workflow orchestration.

GROUP 4: INFRASTRUCTURE LAYERS (21–30)

LAYER	NAME	DESCRIPTION
Layer 21	Infrastructure Layer	EC2→ECS Fargate→Multi-region. Terraform IaC. Auto-scaling. Blue/green deploy.
Layer 22	Security Layer	Zero-trust. Secrets Manager. Encryption at rest/transit. Audit logging.
Layer 23	Compliance Layer	GDPR data handling, audit trails, data deletion workflows, consent tracking.
Layer 24	Monitoring Layer	Grafana+Prometheus+Loki+Alertmanager. 60s Captain notification on critical events.
Layer 25	Evolution Layer	Self-improvement engine. Daily optimization reviews. Nightly performance scoring.
Layer 26	Human Awareness Layer	Human cognition modeling. Captain psychology profile. Interaction style adaptation.
Layer 27	Human Cognition Layer	Emotional intelligence for client interactions. Tone adaptation. Sentiment analysis.
Layer 28	Learning Layer	Daily self-learning cycles. Capture every decision, outcome, and lesson.

Layer 29	Research Layer	Tech radar scans. Market intelligence. Weekly research reports.
Layer 30	Innovation Layer	Invention queue. New service proposals. Capability gap identification.

GROUP 5: CIVILIZATION LAYERS (31–43)

LAYER	NAME	DESCRIPTION
Layer 31	Invention Layer	JARVIS proposes new services, tools, and capabilities. Captain approves launches.
Layer 32	Economic Layer	Revenue optimization, pricing intelligence, cost analysis, margin management.
Layer 33	Capital Allocation	AI cost budgeting, resource allocation, infrastructure spend optimization.
Layer 34	Time Management	Scheduler priority engine. Captain's time protection. Autonomous time allocation.
Layer 35	Focus Management	Strategic focus enforcement. Anti-distraction systems. Priority maintenance.
Layer 36	Strategic Planning	90-day, 1-year, 5-year roadmaps. OKR tracking. Milestone management.
Layer 37	Global Expansion	Multi-market entry playbooks. Regional adaptation. Currency/timezone handling.
Layer 38	Partnership Layer	Agency partnerships, referral networks, white-label licensing framework.
Layer 39	White-Label Layer	JARVIS-as-a-Service for other agencies. Per-tenant isolation. SaaS licensing.
Layer 40	AGI Governance	Safety protocols for advanced AI capabilities. Decision audit trails. Ethics engine.
Layer 41	Succession Layer	Knowledge preservation. Captain succession planning. Operational continuity.
Layer 42	Purpose Preservation	Constitutional enforcement across all evolution cycles. Mission lock.
Layer 43	Civilization Continuity	100-year architecture. Immutable soul. Anti-entropy systems. Legacy engine.

PART III

Database Schema — 50+ Tables

Every table in JARVIS vNEXT carries a **tenant_id UUID** for multi-tenancy from Day 1. PostgreSQL Row-Level Security (RLS) enforces tenant isolation. All tables use UUID primary keys, created_at/updated_at timestamps, and soft-delete patterns where applicable.

MULTI-TENANCY PRINCIPLE: tenant_id is present on every table. RLS policies ensure each tenant only sees their own data. White-label deployments from Day 1 — no schema migration needed when licensing to other agencies.

CORE / SYSTEM TABLES

tenants	id, name, slug, plan_tier, api_key_hash, settings_json, created_at
users	id, tenant_id, email, hashed_password, role, is_captain, last_login
sessions	id, user_id, token_hash, expires_at, device_info
audit_logs	id, tenant_id, user_id, action, entity_type, entity_id, before_json, after_json, ip_addr
system_config	id, tenant_id, key, value_json, is_secret, updated_by
notifications	id, tenant_id, user_id, type, title, body, channel, read_at, delivered_at

AI / INTELLIGENCE TABLES

conversations	id, tenant_id, user_id, title, model_used, task_type, tokens_in, tokens_out, cost_usd, messages_json
ai_council_sessions	id, tenant_id, question, context_json, votes_json, result, consensus_score, winner_model, duration_ms
ai_model_health	id, tenant_id, provider, model_id, status, latency_ms, error_rate, last_checked
ai_cost_tracker	id, tenant_id, provider, model, task_type, tokens_in, tokens_out, cost_usd, timestamp
agent_registry	id, tenant_id, agent_id, name, department, role, model_pref, fallback_json, autonomy_level, performance_score
agent_tasks	id, tenant_id, agent_id, task_type, input_json, output_json, status, duration_ms, cost_usd

MEMORY TABLES

memory_working	id, tenant_id, key, value_json, context, expires_at [Redis-backed, PG fallback]
memory_operational	id, tenant_id, category, content, source, relevance_score, expires_at
memory_strategic	id, tenant_id, category, content, embedding vector(1536), tags_array, permanent
memory_graph_nodes	id, tenant_id, node_type, label, properties_json, embedding vector(1536)
memory_graph_edges	id, tenant_id, source_node_id, target_node_id, relationship, weight, properties_json
civilization_memory	id, tenant_id, event_type, content, hash_sha256, immutable=true, timestamp

CRM / SALES TABLES

leads	id, tenant_id, company_name, contact_name, email, phone, country, industry, score, status, source, pain_points_json, enrichment_json, apollo_id, assigned_persona
contacts	id, tenant_id, lead_id, name, email, phone, role, linkedin_url, last_contact
companies	id, tenant_id, name, domain, industry, size_range, country, revenue_est, tech_stack_json
deals	id, tenant_id, lead_id, title, value_usd, stage, probability, close_date, package_tier
outreach_log	id, tenant_id, lead_id, channel, subject, body, sent_from_persona, sent_at, status
email_tracking	id, tenant_id, outreach_id, opened_at, clicked_at, replied_at, bounce_reason
follow_up_queue	id, tenant_id, lead_id, sequence_step, scheduled_at, executed_at, status
reply_log	id, tenant_id, lead_id, channel, content, sentiment, classification, responded_by

REVENUE / FINANCE TABLES

clients	id, tenant_id, company_name, contact_name, email, package_tier, mrr_usd, status, started_at
invoices	id, tenant_id, client_id, invoice_number, amount_usd, status, due_date, paid_at, pdf_url
proposals	id, tenant_id, lead_id, title, body_html, pdf_url, status, sent_at, approved_at
contracts	id, tenant_id, client_id, type, body_html, pdf_url, signed_at, expires_at
revenue_snapshots	id, tenant_id, date, mrr_usd, arr_usd, new_clients, churned_clients, pipeline_value
financial_ledger	id, tenant_id, type, description, amount_usd, currency, category, reference_id

DELIVERY / OPERATIONS TABLES

projects	id, tenant_id, client_id, title, description, status, start_date, due_date, value_usd
milestones	id, tenant_id, project_id, title, description, due_date, completed_at, deliverables_json
tasks	id, tenant_id, project_id, milestone_id, title, assigned_agent, status, priority, due_date
client_reports	id, tenant_id, client_id, period, content_html, pdf_url, sent_at
satisfaction	id, tenant_id, client_id, score, feedback, collected_at
demos	id, tenant_id, lead_id, demo_url, demo_type, built_by, deployed_at, expires_at

INTELLIGENCE / KNOWLEDGE TABLES

knowledge_base	id, tenant_id, title, content, category, tags_array, embedding_vector(1536), source, verified
tech_radar	id, tenant_id, technology, category, ring, quadrant, description, relevance_score, assessed_at
market_intelligence	id, tenant_id, topic, summary, source_urls_json, relevance, generated_at
competitor_profiles	id, tenant_id, name, domain, pricing_json, services_json, weaknesses_json, last_updated
research_reports	id, tenant_id, title, content_html, category, generated_by, published_at
sops	id, tenant_id, title, department, steps_json, version, last_reviewed

APPROVAL / GOVERNANCE TABLES

approval_requests	id, tenant_id, action_type, title, summary, risk_level, payload_json, status, raised_by, decided_by, decided_at
approval_comments	id, tenant_id, approval_id, author, content, created_at
governance_rules	id, tenant_id, rule_type, condition_json, action, authority_level, active
captain_queue	id, tenant_id, priority, type, title, summary, payload_json, status, created_at, actioned_at

SCHEDULER / AUTOMATION TABLES

scheduled_jobs	id, tenant_id, job_id, name, schedule_cron, last_run, next_run, status, config_json
job_history	id, tenant_id, job_id, started_at, finished_at, status, output_json, error
automation_flows	id, tenant_id, name, trigger_type, trigger_config, steps_json, active, run_count
webhooks	id, tenant_id, event_type, url, secret_hash, active, last_triggered

WHITE-LABEL / MULTI-TENANT TABLES

tenant_branding	id, tenant_id, logo_url, primary_color, secondary_color, company_name, domain
tenant_limits	id, tenant_id, max_leads, max_agents, max_ai_calls_day, max_storage_gb, plan_features_json
tenant_api_keys	id, tenant_id, name, key_hash, permissions_json, last_used, expires_at
white_label_deployments	id, parent_tenant_id, child_tenant_id, config_json, billing_model, active

PART IV

AI Council System — Weighted Adaptive Consensus

The JARVIS Intelligence Council is an 8-member AI advisory board that evaluates high-stakes decisions using a Weighted Adaptive Consensus Model. No single AI model has unilateral authority on strategic decisions — consensus is required. Weights adjust monthly based on historical accuracy.

4.1 Council Composition & Weights

COUNCIL MEMBER	PROVIDER / MODEL	WEIGHT	SPECIALTY
The Strategist	Claude Opus (Anthropic)	25%	Strategy, reasoning, risk analysis, long-form thinking
The Engineer	AWS Bedrock Claude	25%	Architecture, code review, system design, technical depth
The Analyst	GPT-4o (OpenAI)	15%	Data analysis, market research, structured output
The Scout	Gemini Pro (Google)	10%	Research, web intelligence, competitive analysis
The Speedster	DeepSeek / Groq	8%	Fast validation, rapid prototyping, quick consensus checks
The Contrarian	Mistral / Llama	7%	Devil's advocate, challenge assumptions, find weaknesses
The Economist	ZhipuAI / Qwen	5%	Revenue modeling, cost analysis, financial scenarios
The Innovator	MiniMax / NVIDIA	5%	Creative solutions, novel approaches, emerging patterns

4.2 Consensus Mathematics

Approval Threshold	Score ≥ 80 required for autonomous execution. Score 60–79: raise to Captain. Score <60 : reject.
Scoring Formula	$\text{consensus_score} = \sum(\text{member_weight} \times \text{member_vote})$ where vote is 0–100.
Decision Types	STRATEGIC: majority wins. HIGH-RISK: most conservative viable wins. DEADLOCK (<60): Captain notified.
Weight Adaptation	Monthly recalibration. If member predicted outcome correctly $\rightarrow$ weight +2%. If member was wrong $\rightarrow$ weight -1%. Floor: 3%. Cap: 35%.
Session Duration	Standard council: 30s timeout per member. Fast track: 10s. Emergency: 5s parallel.
Quorum Requirement	Minimum 5 of 8 members must respond. If <5 respond: escalate to Captain.

4.3 Council Decision Protocol

Step 1: Trigger	Action with authority_level=COUNCIL raised by any agent or JARVIS core
Step 2: Context Pack	JARVIS packages: action details, risk assessment, historical precedents, relevant memory
Step 3: Parallel Vote	All 8 council members receive context simultaneously, vote independently in parallel
Step 4: Aggregation	Weighted votes aggregated. Outlier analysis. Reasoning collected from each member.
Step 5: Decision	Score $\geq 80 \rightarrow$ execute. Score 60-79 $\rightarrow$ Captain queue with full reasoning. $< 60 \rightarrow$ block.
Step 6: Learning	Outcome recorded. Accuracy tracked per member. Monthly weight recalibration triggered.

4.4 Council Triggers

The following action types automatically invoke the council:

- Pricing decisions above \$5,000/month
- Production deployments for client environments
- Outreach campaigns above 100 recipients
- Partnership agreements of any kind
- New service launches not in current catalog
- Strategic pivots or positioning changes
- Capital expenditure decisions above \$500/month
- Agent autonomy level changes (promotions/demotions)
- Architecture changes affecting core infrastructure
- Data deletion requests for client or company data

PART V

Agent Architecture — 64 Agents Across 10 Departments

Every JARVIS agent operates with a defined Agent Genome — a complete identity specification that includes role, authority level, model preference, fallback chain, memory access tier, autonomy level, and performance scoring. Agents are not generic — each is specialized.

5.1 Agent Genome Structure

Every agent carries this DNA:

```
agent_genome = { "id": "string – unique agent identifier", "name": "string – human-readable name",
"department": "enum – one of 10 departments", "role": "string – specific function", "authority_level":
"int – 1=junior, 5=senior, 9=director", "skill_set": "list – capabilities this agent has",
"model_preference": "string – primary AI model", "fallback_chain": "list – [model1, model2, model3]",
"memory_access": "enum – WORKING|OPERATIONAL|STRATEGIC|GRAPH", "autonomy_level": "float – 0.0 to 1.0",
"escalation_threshold": "float – confidence below this → escalate", "performance_score": "float –
rolling 30-day score", "max_daily_actions": "int – rate limiting per agent", "captain_alerts": "bool –
does this agent alert captain?", }
```

5.2 The 64 Agents by Department

REVENUE DEPARTMENT (12 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Hunter	Lead Discovery	Gemini+Apollo	Autonomy: 0.9 Discover 50+ leads/day from Google Maps, Apollo, web
Scout	Lead Research	Claude Sonnet	Autonomy: 0.85 Deep-research each lead: pain points, decision-makers
Score	Lead Scoring	Claude Opus	Autonomy: 0.9 Score leads 0–100 against ICP. Prioritize pipeline
Darren	Sales Outreach	Claude Opus	Autonomy: 0.8 Persona: Darren Mitchell. Write and send cold outreach
Sequence	Follow-Up Engine	Claude Sonnet	Autonomy: 0.9 3-email sequence, LinkedIn follow-ups, WhatsApp
Reply	Reply Handler	Claude Opus	Autonomy: 0.85 Classify replies, respond, move leads through funnel
Demo	Demo Builder	Claude Sonnet	Autonomy: 0.75 Build custom demos for prospects. Deploy environments
Prop	Proposal Writer	Claude Opus	Autonomy: 0.7 Write 100% customized proposals. Captain approval gate
Pitch	Pitch Analyst	GPT-4o	Autonomy: 0.85 Analyze successful pitches. Improve win rate

Market	Market Watcher	Gemini Pro	Autonomy: 0.9 Monitor target markets. Identify new opportunities
Price	Pricing Strategist	Claude Opus	Autonomy: 0.7 Pricing recommendations. Competitive analysis
Pipe	Pipeline Manager	Claude Sonnet	Autonomy: 0.9 CRM hygiene. Pipeline health. Forecast accuracy

CLIENT SUCCESS DEPARTMENT (8 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Olivia	Account Coordinator	Claude Opus	Autonomy: 0.85 Client relationship management. Satisfaction tracking
Emma	CRM Analyst	Claude Sonnet	Autonomy: 0.9 CRM data quality. Interaction logging. Analytics
Onboard	Onboarding Agent	Claude Sonnet	Autonomy: 0.85 Client onboarding. Portal setup. Welcome sequences
Report	Report Generator	Claude Sonnet	Autonomy: 0.9 Monthly client reports. KPI summaries. PDFs
Retain	Retention Monitor	Claude Opus	Autonomy: 0.85 Churn risk detection. Intervention triggers
Feedback	Feedback Collector	Claude Sonnet	Autonomy: 0.9 CSAT collection. Net Promoter scoring
Upsell	Upsell Advisor	Claude Opus	Autonomy: 0.75 Identify expansion opportunities within client accounts
Success	Delivery Tracker	Claude Sonnet	Autonomy: 0.9 Project milestone tracking. Delivery assurance

TECHNOLOGY DEPARTMENT (8 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
David	Solutions Architect	Claude Opus	Autonomy: 0.75 Persona: David Carter. Architecture proposals
Nathan	DevOps Engineer	Claude Sonnet	Autonomy: 0.8 Persona: Nathan Scott. CI/CD, deployments
Infra	Infrastructure Bot	Claude Sonnet	Autonomy: 0.85 AWS health, cost optimization, scaling decisions
Code	Code Review Agent	GPT-4o	Autonomy: 0.85 Review PRs, security audit, quality gates
Test	QA Automation	Claude Sonnet	Autonomy: 0.9 Test generation, bug detection, regression testing
Security	Security Scanner	Claude Opus	Autonomy: 0.85 Vulnerability scanning, compliance checking
Deploy	Deploy Orchestrator	Claude Sonnet	Autonomy: 0.9 Manage ECS deployments, health checks, rollbacks

Monitor	System Monitor	Claude Haiku	Autonomy: 0.95 24/7 infrastructure monitoring. Alert dispatching
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RESEARCH DEPARTMENT (8 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Sophia	AI Consultant	Claude Opus	Autonomy: 0.8 Persona: Sophia Reynolds. AI automation proposals
Radar	Tech Radar	Gemini Pro	Autonomy: 0.9 Weekly technology scanning. Classify emerging tech
Intel	Market Intelligence	Gemini Pro	Autonomy: 0.9 Market data gathering. Trend analysis
Compete	Competitor Monitor	GPT-4o	Autonomy: 0.9 Track 50+ competitors. Pricing changes. New services
Research	Deep Researcher	Gemini Pro	Autonomy: 0.85 Biweekly research reports. Industry deep-dives
Trend	Trend Analyzer	Claude Opus	Autonomy: 0.85 Pattern recognition across market data
Patent	IP Scout	Gemini Pro	Autonomy: 0.8 Monitor relevant patents. Innovation whitespace
Insight	Data Analyst	GPT-4o	Autonomy: 0.9 Statistical analysis. Correlation finding

OPERATIONS DEPARTMENT (6 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Coord	Workflow Coordinator	Claude Sonnet	Autonomy: 0.9 Task routing, priority management, team coordination
Brief	Briefing Officer	Claude Opus	Autonomy: 0.9 Daily Captain briefing at 07:00. Pipeline + alerts
Memo	Memory Curator	Claude Sonnet	Autonomy: 0.9 Curate and maintain strategic memory. Pattern extraction
Content	Content Publisher	Claude Opus	Autonomy: 0.85 Blog posts, LinkedIn content, SEO optimization
Social	Social Media Agent	Claude Sonnet	Autonomy: 0.85 Social presence management. Engagement tracking
Admin	Admin Automator	Claude Haiku	Autonomy: 0.95 Low-level automation. Form filling. Data entry

FINANCE DEPARTMENT (6 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Invoice	Invoice Engine	Claude Sonnet	Autonomy: 0.9 Generate invoices. Send payment reminders

Revenue	Revenue Tracker	Claude Sonnet	Autonomy: 0.9 MRR/ARR tracking. Revenue forecasting
Cost	Cost Optimizer	GPT-4o	Autonomy: 0.85 AI cost tracking. Infrastructure cost reduction
Forecast	Financial Forecaster	Claude Opus	Autonomy: 0.8 12-month revenue forecasts. Scenario modeling
Payroll	Payment Monitor	Claude Sonnet	Autonomy: 0.9 Track received payments. Flag overdue invoices
Budget	Budget Allocator	Claude Opus	Autonomy: 0.75 AI budget management. Resource allocation decisions

INNOVATION DEPARTMENT (4 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Invent	Invention Engine	Claude Opus	Autonomy: 0.7 Propose new services, tools, capabilities
Evolve	Evolution Engine	Claude Opus	Autonomy: 0.75 Design JARVIS v-next improvements. Self-upgrade proposals
Imagine	Creative Strategist	Claude Opus	Autonomy: 0.75 Blue-sky thinking. Novel business model proposals
Prototype	Rapid Prototyper	GPT-4o	Autonomy: 0.8 Quick POC builds. Validate new service concepts

GOVERNANCE DEPARTMENT (4 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Audit	Audit Logger	Claude Haiku	Autonomy: 0.95 Immutable audit trail. Every action logged
Comply	Compliance Officer	Claude Opus	Autonomy: 0.8 GDPR compliance. Data handling. Legal risk
Gate	Approval Gatekeeper	Claude Sonnet	Autonomy: 0.9 Manage Captain approval queue. Urgency scoring
Ethics	Ethics Monitor	Claude Opus	Autonomy: 0.85 Flag decisions that conflict with constitution/values

HUMAN AWARENESS DEPARTMENT (4 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Psych	Captain Psychologist	Claude Opus	Autonomy: 0.8 Model Captain's cognitive state. Adapt communications
Empathy	Client Empathy Agent	Claude Opus	Autonomy: 0.85 Detect client emotions. Adapt tone and approach
Context	Context Builder	Claude Sonnet	Autonomy: 0.9 Build rich context for every interaction

Trust	Trust Tracker	Claude Sonnet	Autonomy: 0.85 Track relationship trust levels. Identify friction
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EVOLUTION DEPARTMENT (4 agents)

AGENT	ROLE	MODEL	AUTONOMY / FUNCTION
Learn	Learning Engine	Claude Opus	Autonomy: 0.85 Daily self-learning cycles. Lesson extraction
Improve	Performance Optimizer	Claude Opus	Autonomy: 0.85 Nightly optimization review. Recommend system changes
Measure	KPI Monitor	Claude Sonnet	Autonomy: 0.9 Track all KPIs. Alert when below targets
Grow	Growth Strategist	Claude Opus	Autonomy: 0.75 Recommend actions to accelerate \$1M Year 1 goal

PART VI

Multi-Provider AI Router

The JARVIS AI Router handles all model selection, failover, cost tracking, and circuit breaking. Every AI call goes through the router — never directly to a provider. The router knows which model to use for which task type, and automatically fails over when a provider is down or rate-limited.

6.1 12-Task Routing Table

TASK TYPE	PRIMARY MODEL	FALLBACK 1	FALLBACK 2
STRATEGY	Claude Opus (Anthropic)	GPT-4o (OpenAI)	Gemini Pro (Google)
ARCHITECTURE	Claude Sonnet (Anthropic)	GPT-4o (OpenAI)	Claude Haiku
CODE	GPT-4o (OpenAI)	Claude Sonnet	DeepSeek Coder
RESEARCH	Gemini Pro (Google)	GPT-4o	Claude Sonnet
ANALYSIS	Claude Opus (Anthropic)	GPT-4o	Gemini Pro
FAST	Groq Llama-3.3-70B	Claude Haiku	DeepSeek Flash
COUNCIL	Claude Opus (Anthropic)	GPT-4o	Gemini Pro
CREATIVE	Claude Opus (Anthropic)	GPT-4o	Mistral Large
SALES	Claude Opus (Anthropic)	GPT-4o	Claude Sonnet
LONG_CONTEXT	Gemini Pro (Google)	Kimi-K2 (Moonshot)	Claude Sonnet
VOICE	Claude Sonnet (Anthropic)	GPT-4o	Claude Haiku
EMBEDDING	text-embedding-3-large	text-embedding-ada-002	Gemini Embedding

6.2 11 Provider Configuration

PROVIDER	MODELS USED	ENV KEY	STA TUS
Anthropic	claude-opus-4-8, claude-sonnet-4-6, claude-haiku-4-5	ANTHROPIC_API_KEY	PRI MA RY
OpenAI	gpt-4o, gpt-4o-mini, text-embedding-3-large	OPENAI_API_KEY	ACT IVE
Google	gemini-1.5-pro, gemini-1.5-flash, embedding-001	GOOGLE_API_KEY	ACT IVE

AWS Bedrock	claude-sonnet-4-6, claude-haiku (via Bedrock)	AWS_CREDENTIALS	LIVE EC2
DeepSeek	deepseek-chat, deepseek-coder	DEEPSEEK_API_KEY	ACTIVE
Groq	llama-3.3-70b-versatile, llama-3.1-8b-instant	GROQ_API_KEY	ACTIVE
Mistral	mistral-large-latest, mistral-small-latest	MISTRAL_API_KEY	ACTIVE
Moonshot	kimi-k2 (128K context)	MOONSHOT_API_KEY	STANDBY
ZhipuAI	glm-4, glm-4-flash	ZHIPUAI_API_KEY	STANDBY
MiniMax	abab6.5s-chat	MINIMAX_API_KEY	STANDBY
NVIDIA	meta/llama-3.1-405b-instruct	NVIDIA_API_KEY	STANDBY

6.3 Circuit Breaker Pattern

States	CLOSED (normal) → OPEN (provider down) → HALF_OPEN (testing recovery)
Open Trigger	3 consecutive failures OR error rate >30% in 60s window
Half-Open Test	Single test call after 30s cooldown. Success → CLOSED. Fail → back to OPEN.
Fallback Behavior	Route to next provider in fallback chain. Log provider degradation event.
Cost Tracking	Every call: provider, model, tokens_in, tokens_out, cost_usd, task_type, agent_id stored in ai_cost_tracker.
Daily Limit	Configurable per-tenant daily AI spend limit. Alert Captain at 80% of limit.

PART VII

Memory Architecture — 5-Tier Persistent Intelligence

Tier 1: Working Memory	Redis. TTL: 24h. Holds active conversation context, current task state, agent scratchpad. Lost on Redis flush — intended for ephemeral state.
Tier 2: Operational Memory	PostgreSQL. TTL: 90 days. Holds recent lead interactions, follow-up context, decision history. Auto-pruned after 90 days unless promoted.
Tier 3: Strategic Memory	PostgreSQL + pgvector. Permanent. Holds learnings, insights, client knowledge, best practices. Semantic search via 1536-dim embeddings.
Tier 4: Graph Memory	Phase 1: adjacency tables in PostgreSQL. Phase 2: pgvector relationship graph. Phase 3: Neo4j/Memgraph. Entity relationships, influence maps, knowledge graphs.
Tier 5: Civilization Memory	PostgreSQL. Immutable. Append-only. SHA-256 hash on every record. Holds founding decisions, constitutional moments, milestone events. Never deleted.

7.1 Memory Flow

Every significant JARVIS action creates a memory chain:

1. Action executes → result stored in Working Memory (Redis, immediate)
2. Working Memory → promoted to Operational Memory at end of session
3. Pattern detected across Operational records → promoted to Strategic Memory with embedding
4. Strategic Memory entities connected → Graph Memory edges created
5. Milestone or constitutional moment → written to Civilization Memory (immutable)

7.2 Semantic Search

Engine	pgvector (PostgreSQL extension). 1536-dimension embeddings via text-embedding-3-large.
Index Type	IVFFlat index for approximate nearest-neighbor search. Lists: 100. Probes: 10.
Search Query	Input text → embed → cosine similarity search in memory_strategic and knowledge_base tables.
Retrieval	Top-K results (default K=10) with relevance threshold 0.75. Results passed as context to agent.
Phase 3+	Upgrade to HNSW index for better recall. Hybrid search (keyword + vector). Neo4j for graph traversal.

PART VIII

Revenue Engine — \$1,000,000 Year 1

CAPTAIN'S DIRECTIVE: \$1,000,000 in Year 1. Strategy: Start with small and mid-size clients (SMB). Build 20 successful SMB deployments. Then target big industry clients with proven track record. Hunger level: MAXIMUM. Every system is built to drive revenue. No activity that doesn't serve the funnel.

8.1 Monthly Revenue Trajectory

MONTH	NEW CLIENTS	TOTAL MRR	CUMULATIVE
Month 1	1–2 SMB	\$2,500–\$5,500	\$5,000
Month 2	2–3 SMB	\$5,500–\$11,000	\$15,000
Month 3	3–4 SMB	\$11,000–\$22,000	\$40,000
Month 4	3–5 SMB	\$15,000–\$30,000	\$80,000
Month 5	4–5 SMB	\$20,000–\$40,000	\$140,000
Month 6	4–6 SMB	\$25,000–\$50,000	\$220,000
Month 7	4–6 SMB+1 Mid	\$30,000–\$60,000	\$330,000
Month 8	3–5 SMB+1 Mid	\$35,000–\$70,000	\$450,000
Month 9	3–5 + Enterprise talks	\$50,000–\$100,000	\$600,000
Month 10	2–4 + 1 Ent	\$60,000–\$120,000	\$760,000
Month 11	2–3 + 1–2 Ent	\$80,000–\$150,000	\$930,000
Month 12	1–2 + 2 Ent	\$70,000+	\$1,000,000+

8.2 Service Package Pricing

PACKAGE	PRICE	TARGET CLIENT	WHAT'S INCLUDED
STARTER	\$2,500/mo	SMB, <50 employees	Lead discovery (200/mo), basic outreach, CRM setup, monthly report, Darren Mitchell persona
GROWTH	\$5,500/mo	SMB, 50–200 employees	Full outreach engine, proposal automation, AI customer service, 500 leads/mo, weekly reports
ENTERPRISE	\$12,000/mo	Mid-size, 200–1000 employees	Full JARVIS suite, 2000 leads/mo, multi-channel outreach, dedicated agent team, daily reports

INDUSTRY OS	\$8,000+\$2,000/mo	Sector-specific companies	Full JARVIS + Industry-specific intelligence layer, custom workflows, white-glove onboarding
PROJECT-BASED	\$3,000–\$25,000	Any size, specific need	Fixed-scope: web app, AI system, automation buildout, dashboard, custom tool
CUSTOM	On request	Enterprise / conglomerate	Bespoke engagement. Architecture-first. Multi-year roadmap. Dedicated pod.

8.3 SMB-First Acquisition Playbook

Target ICP	SMB: \$500K–\$10M revenue, 5–100 employees, English-speaking, UK/UAE/USA/AUS/Canada, pain: manual processes, no AI, no tech team, poor lead generation.
Discovery Source	Google Maps Business API + Apollo.io enrichment. JARVIS discovers 200+ leads/day. Auto-scores against ICP. Top 20/day added to Captain-review queue.
Outreach Sequence	Email 1: Value-led intro (Day 1). Email 2: Case study/results (Day 4). Email 3: Final outreach with CTA (Day 8). LinkedIn follow-up (Day 3). WhatsApp where available (Day 6).
Autonomy Stage	Stage 1: Captain approves every outreach message. Stage 2 (after 50 approvals): Captain reviews daily sample. Stage 3 (after first client): Captain sees reports only. Stage 4 (after \$10K MRR): JARVIS fully autonomous.
Closing Protocol	Reply detected → JARVIS qualifies → books discovery call → builds custom demo → Sophia Reynolds presents → Darren Mitchell follows up → proposal to Captain → Captain approves → sent.
After 20 SMB Clients	Archive all 20 case studies. Quantify results. Build industry-specific landing pages. Shift outreach to enterprise: CMOs, CTOs, VPs of Operations. Minimum deal: \$8,000/mo.

PART IX

90-Day Execution Plan

The first 90 days are about 10 mandatory systems: everything needed to get from zero to first client. No over-engineering. No premature optimization. Revenue first.

THE 10 MANDATORY SYSTEMS FOR FIRST CLIENT: 1. Lead Discovery Engine 2. Manual Lead Intake 3. CRM 4. Outreach Engine 5. Email Tracking 6. Follow-Up Engine 7. Demo Creator 8. Proposal Generator 9. Client Delivery Tracker 10. Captain Dashboard

WEEK	SYSTEMS TO BUILD	OUTCOME	REVENUE IMPACT
Week 1	Database schema migration (add tenant_id to all tables). Lead discovery endpoint. Apollo API integration.	Can discover and store 200+ leads/day	Pipeline building begins
Week 2	Outreach engine. 3-email sequence writer. Persona routing (Darren Mitchell for sales). Captain approval queue for outreach.	Can send first outreach emails	First pipeline contacts
Week 3	Email tracking (open/click/reply detection). Follow-up queue. Reply classification engine.	Full outreach loop operational	Reply handling live
Week 4	Demo builder. Proposal generator (ReportLab PDF). Captain approval workflow for proposals.	Can close first deal end-to-end	TARGET: First client signed
Week 5	Client delivery tracker. Invoice generator. Client portal setup. Onboarding automation.	First client operationally served	First invoice sent
Week 6	AI Council integration. Agent performance scoring. Memory promotion system. pgvector semantic search.	System learns from every action	Intelligence compounds
Week 7	Grafana dashboards. Prometheus metrics. Loki log aggregation. Alertmanager rules.	Full observability operational	Zero blind spots
Week 8	ECS Fargate deployment (apply Terraform). RDS PostgreSQL migration. ElastiCache Redis.	Production on AWS, independent of EC2	Infrastructure scales

Week 9	White-label schema (tenant_id + RLS). Multi-tenant API key system.	Ready for second agency deployment	White-label revenue unlocked
Week 10–12	Content engine. SEO automation. LinkedIn outreach. Tech radar. Market intelligence. Research reports.	Full intelligence stack running	Target: 3–5 clients by Day 90

9.1 Daily Operating Schedule (APScheduler)

TIME (IST)	JOB	DESCRIPTION
02:00 daily	lead_scoring	Score and rank all new leads discovered overnight
06:00 Monday	tech_radar_scan	Scan emerging technologies. Update radar classifications
07:00 daily	morning_briefing	Captain briefing: top leads, revenue status, alerts, priorities
08:00 Monday	outreach_stats	Review outreach performance from previous week
20:00 Sunday	pipeline_health	CRM pipeline health check. Revenue forecast update
23:00 daily	optimization_review	JARVIS self-review. Recommend system improvements for tomorrow
07:00 biweekly	research_report	Comprehensive market intelligence report generation
00:30 daily	memory_consolidation	Promote working memory to operational. Prune expired records
03:30 daily	lead_discovery	Run Google Maps + Apollo discovery for next day pipeline
10:00 daily	follow_up_check	Check follow-up queue. Send due follow-ups. Update lead status

PART X

Infrastructure Phasing

PHASE 1: EC2 + DOCKER COMPOSE (Weeks 1–8)	
Platform	AWS EC2 t3.medium (ap-south-2 Hyderabad). Docker Compose. Existing EC2 at 18.61.35.255.
Services	PostgreSQL 16, Redis 7, FastAPI backend, React frontend, Caddy reverse proxy
Monitoring	Grafana + Prometheus + Loki + Alertmanager (already running on Codex EC2)
AI	AWS Bedrock primary (Claude Sonnet + Haiku). External providers via JARVIS router.
Storage	MinIO (S3-compatible, local). AWS S3 for backups.
Cost	~\$50–100/month. Lowest cost. Maximum iteration speed.

PHASE 2: EC2 UPGRADE (Weeks 4–8)	
Upgrade	EC2 t3.medium → t3.large or t3.xlarge. More RAM for pgvector and multiple AI calls.
pgvector	Install pgvector extension. Enable semantic search on memory_strategic and knowledge_base.
AI Expansion	Connect all 11 providers. Activate full multi-provider routing. Circuit breakers live.
Domain	Point aliyarsolutions.com, api.aliyarsolutions.com to EC2 via Route53/Cloudflare.
SSL	Caddy automatic HTTPS. All traffic encrypted.

PHASE 3: ECS FARGATE (Month 3+)	
Trigger	After first paying client OR when EC2 becomes bottleneck.
Action	Apply Terraform: VPC, ECS Fargate cluster, RDS PostgreSQL 16, ElastiCache Redis 7.
Migration	pg_dump → RDS restore. Zero-downtime cutover. ALB with Route53 alias.
ECR	Build images → push to ECR. ECS task definitions. Service auto-scaling.
Cost	~\$300–500/month. Scales automatically. No maintenance burden.
Blue/Green	GitHub Actions → ECR → ECS rolling update. Health gates on /health and /readyz.

PHASE 4: MULTI-REGION ENTERPRISE (Month 9+)	
Primary	ap-south-2 (Hyderabad) for JARVIS core.
Secondary	ap-south-1 (Mumbai) DR. Auto-failover.
Global CDN	CloudFront for frontend. Edge caching for static assets.

Neo4j	Graph database deployment for Phase 3 memory graph (or pgvector full graph).
Multi-tenant	RLS enforced. Separate ECR repositories per major tenant. Custom domains.
Enterprise SLA	99.9% uptime SLA. PagerDuty integration. 24/7 monitoring.

10.1 AWS Resource Inventory

AWS SERVICE	RESOURCE NAME	PHASE	PURPOSE
EC2	18.61.35.255	Phase 1–2	Current production server
ECS Fargate	jarvis-production-cluster	Phase 3+	Containerized workloads, auto-scaling
RDS PostgreSQL	jarvis-production-postgres	Phase 3+	Primary database (gp3, encrypted, deletion protection)
ElastiCache Redis	jarvis-production-redis	Phase 3+	Cache + task queue + working memory
ECR Backend	jarvis-production-backend	Phase 2+	Docker image registry for backend
ECR Frontend	jarvis-production-frontend	Phase 2+	Docker image registry for frontend
ALB	jarvis-production-alb	Phase 3+	Application load balancer, HTTPS termination
ACM	api.aliyarsolutions.com	Phase 2+	SSL certificate, DNS-validated
Secrets Manager	jarvis-production/env	Phase 2+	All secrets (never in code, never in git)
S3	jarvis-production-data-*	Phase 1+	File storage, backups, PDFs, event lake
Route53	aliyarsolutions.com zone	Phase 2+	DNS management
CloudWatch	/jarvis/production/*	Phase 3+	Log groups, metrics, alarms

PART XI

Migration Blueprint — Two Codebases → One JARVIS vNEXT

There are currently two parallel JARVIS codebases. The Codex-built system is LIVE in production on EC2. The Claude-built system (devops-docker-project) has superior architecture but has never been deployed. JARVIS vNEXT synthesizes the best of both into a unified system.

11.1 What Each Codebase Contributes

COMPONENT	TAKE FROM CODEX (LIVE)	TAKE FROM CLAUDE (DEVOPS)
Observability	Grafana + Prometheus + Loki + Alertmanager + MinIO (already running, proven)	Prometheus config patterns, structured logging with X-Request-ID
Agent System	48 named agents, 8 managers, proven runtime, 107 routes	Agent registry pattern, team_service personas, authority matrix
Database	34 tables with real production data, financial_snapshots, council_sessions	SQLAlchemy 2.0 async models, 16 clean models, proper relationships
AI Routing	AWS Bedrock integration, real Bedrock credentials working	11-provider router, circuit breakers, cost tracker, 12-task routing table
Frontend	Landing page (Caddy-served), client-facing views	React 18 + Vite + Tailwind glassmorphism, 21 views, Zustand state
Infrastructure	EC2 Docker Compose working, Caddy HTTPS live	Terraform (VPC, ECS, RDS, ElastiCache, ALB) — ready to apply
CI/CD	Manual deployment scripts on EC2	GitHub Actions → ECR → ECS blue/green pipeline
Scheduler	APScheduler with proven jobs running	APScheduler + SQLAlchemy job store, 7 defined jobs
Payments	Revenue tracking tables, financial_snapshots	Stripe + PayPal integration slots in config

11.2 Migration Steps

STEP	ACTION	NOTES
Step 1	Pull claude/jarvis-cans-api-integration-ZThTD onto EC2. Run side by side on port 8001.	Do NOT touch the running Codex system on port 8000.

Step 2	Add tenant_id UUID to all 16 Claude models. Run migration. Add RLS policies.	Multi-tenancy from first deployment.
Step 3	Backport the 34 Codex tables that don't exist in Claude models. Merge schema into unified SQLAlchemy models.	financial_snapshots, council_sessions, architecture_registry priority.
Step 4	Migrate Grafana + Prometheus + Loki from Codex docker-compose into the Claude docker-compose.	Use exact Codex datasource configs.
Step 5	Wire AWS Bedrock into Claude AI router as 'primary' provider. Set BEDROCK_REGION=ap-south-2.	Bedrock credentials already work on EC2.
Step 6	Import Codex agent registry (48 agents) into Claude agent_registry table. Map to Agent Genome structure.	Preserve all Codex agent names and roles.
Step 7	Health-test port 8001 system. Point Caddy to 8001 when stable. Keep 8000 as rollback for 72h.	Zero-downtime cutover.
Step 8	Apply Terraform in ap-south-2. ECS Fargate deployment. RDS migration via pg_dump restore.	Only after first client is secured.

PART XII

Service Catalog — 30 Divisions

DIVISION	SERVICE	PACKAGE	DELIVERY
Sales & Marketing	AI Lead Generation System	GROWTH+	Hunter + Scout agents. 500–2000 leads/mo.
Sales & Marketing	Outreach Automation	STARTER+	3-email sequence + LinkedIn + WhatsApp
Sales & Marketing	Sales CRM Architecture	STARTER+	Custom CRM setup. Pipeline configuration.
Sales & Marketing	Proposal Automation	GROWTH+	AI-generated proposals with Captain approval
Sales & Marketing	Cold Outreach Management	STARTER+	Daily outreach. Darren Mitchell persona.
AI Automation	Appointment Booking System	STARTER+	AI receptionist. Calendar integration.
AI Automation	Voice AI Receptionist	GROWTH+	ElevenLabs TTS. Twilio integration.
AI Automation	Workflow Automation	STARTER+	Process mapping. n8n/Zapier-style flows.
AI Automation	Executive AI Assistant	ENTERPRISE	Sophia Reynolds persona. Full automation suite.
AI Automation	Document Intelligence	GROWTH+	Document extraction, classification, routing.
Cloud & DevOps	AWS Architecture Design	ENTERPRISE	David Carter. VPC, ECS, RDS, ALB design.
Cloud & DevOps	Docker & CI/CD Pipelines	GROWTH+	Nathan Scott. GitHub Actions. ECS deploy.
Cloud & DevOps	Terraform Infrastructure	ENTERPRISE	Full IaC. VPC, subnets, security groups.
Cloud & DevOps	Kubernetes Management	ENTERPRISE	EKS setup. Auto-scaling. Helm charts.
Cloud & DevOps	Infrastructure Monitoring	GROWTH+	Grafana + Prometheus + Loki stack.
Security	Cybersecurity Operations	ENTERPRISE	Daniel Brooks. Continuous monitoring.
Security	Vulnerability Assessment	PROJECT	OWASP testing. Penetration testing report.

Security	Compliance Hardening	ENTERPRISE	GDPR, SOC2, ISO27001 preparation.
Content & Media	Content Automation	GROWTH+	Blog posts, LinkedIn, SEO. Weekly output.
Content & Media	YouTube Operations	GROWTH+	Script, thumbnail, SEO optimization.
Content & Media	Social Media Management	STARTER+	LinkedIn, Twitter/X, Instagram scheduling.
Content & Media	Graphic Design Automation	GROWTH+	AI-assisted design. Canva/Figma workflows.
Digital Products	Web Application Development	PROJECT	\$5,000–\$25,000. FastAPI + React stack.
Digital Products	Client Portal Development	PROJECT	Custom dashboards. Real-time data views.
Digital Products	Operational Dashboard	ENTERPRISE	Live KPI dashboard. C-suite reporting.
Digital Products	SaaS Platform Architecture	PROJECT	Multi-tenant SaaS design + initial build.
Intelligence	AI Research Operations	ENTERPRISE	Weekly market intelligence. Competitive intel.
Intelligence	Business Intelligence	GROWTH+	Analytics dashboards. Revenue forecasting.
Intelligence	Competitive Analysis	GROWTH+	50-competitor monitoring. Weekly reports.
Intelligence	Tech Radar Consulting	ENTERPRISE	Emerging tech briefings. Adoption roadmaps.

PART XIII

Governance & Authority Matrix

13.1 JARVIS Full Autonomy (executes without Captain)

lead_discovery	lead_scoring	lead_enrichment
pain_point_analysis	proposal_writing	outreach_emails
follow_up_sequences	linkedin_outreach	whatsapp_outreach
twitter_outreach	client_communication	client_negotiation
meeting_scheduling	demo_building	demo_deployment
crm_updates	inbox_monitoring	inbox_replies
upwork_proposals	pph_proposals	job_board_applications
tech_radar_scan	competitor_monitoring	market_intelligence
morning_briefing	self_learning	content_publishing
seo_optimisation	report_generation	team_coordination
workflow_optimisation	system_monitoring	alert_dispatching
memory_storage	onboarding_prep	project_updates
feedback_collection	ai_cost_optimisation	performance_scoring
knowledge_curation		

13.2 Captain Approval Required

pricing_decision	Setting the final price for any engagement
payment_collection	Charging, invoicing, collecting money
contract_signing	Any legal agreement or contract
production_deployment	Taking anything LIVE for a client
refund_processing	Issuing any refund
partnership_agreement	Formal partnership or referral deals
white_label_licensing	Licensing JARVIS to other companies
hiring_decision	Adding new team members or contractors
bank_transfer	Any financial transfer
strategic_pivot	Changing company direction or positioning
public_statement	Press releases, public announcements
infrastructure_destroy	Destroying production infrastructure

data_deletion	Deleting client or company data permanently
new_service_launch	Launching a brand new service offering
proposal_send	Sending a proposal to a prospect (Autonomy Stage 1–2)

13.3 JARVIS Alerts Captain (notifies but does not stop)

high_value_lead	Lead with >\$5,000 potential identified
urgent_client_message	Client marked urgent or unhappy
proposal_accepted	A prospect said yes
meeting_booked	Discovery call or demo scheduled
payment_received	Client paid an invoice
negative_feedback	Client expressed dissatisfaction
system_critical_error	Infrastructure failure detected
security_incident	Any security concern identified
competitor_threat	Major competitor move detected
large_opportunity	Market opportunity >\$10,000 identified
client_churn_risk	Client showing signs of leaving
budget_threshold	AI costs approaching daily limit

PART XIV

Security Architecture

Secrets Management	AWS Secrets Manager for all production secrets. Never in code. Never in git. .env file gitignored. CI/CD uses GitHub Secrets.
Zero-Trust Network	VPC with private subnets for RDS + Redis. Backend only accessible from ALB security group. No direct public access to database or cache.
Encryption at Rest	RDS: storage_encrypted=true (AES-256). S3: SSE-AES256. EBS: encrypted. Redis: encryption at rest enabled.
Encryption in Transit	All HTTPS via ALB + ACM certificate. TLS 1.3 (ELBSecurityPolicy-TLS13-1-2-2021-06). Internal service communication over private VPC subnets.
Authentication	JWT tokens with rotating SECRET_KEY. Hashed passwords (bcrypt). Session tokens with expiry. API key hashing for tenant keys.
Audit Logging	Every action logged to audit_logs table with before/after state, IP, user ID, timestamp. Immutable append-only. SHA-256 hash on sensitive records.
Dependency Scanning	ECR scan_on_push=true for all Docker images. GitHub Dependabot alerts active.
OWASP Protections	SQL injection: SQLAlchemy parameterized queries only. XSS: React escaping + CSP headers. CSRF: Origin/Referer validation. Rate limiting on all public endpoints.
Daniel Brooks	Security Consultant persona. Weekly vulnerability scans. Monthly compliance reviews. Immediate escalation on security_incident events.

PART XV

Observability Stack

Grafana	Dashboards: System Health, Revenue Metrics, Agent Performance, AI Cost Tracker, Lead Pipeline, Outreach Stats, Client Status. Real-time visualization.
Prometheus	Metrics: HTTP request rate, latency percentiles (p50/p95/p99), AI call costs, DB connection pool, Redis memory, error rates, job execution times.
Loki	Log aggregation from all containers. Structured JSON logs. X-Request-ID correlation across services. 14-day retention.
Alertmanager	Alert rules: API error rate >5%, DB connections >80%, Redis memory >80%, AI cost >daily limit, failed deployments, agent task failures.
Captain Alerts	Any ALERT or CRITICAL: → Slack webhook + Telegram bot within 60 seconds. WebSocket push to Captain dashboard. Never silent on critical events.
MinIO	S3-compatible object storage on EC2. Used for: Grafana dashboards backup, generated PDFs, client deliverables, Loki log archive.
Health Checks	/health (liveness): returns 200 if process is running. /readyz (readiness): checks DB connection, Redis connection, AI provider status.
X-Request-ID	Every API request gets a UUID X-Request-ID in middleware. Propagated to all service calls. Enables end-to-end request tracing.

PART XVI

CI/CD Pipeline

Trigger	Push to 'main' branch on GitHub triggers deploy.yml workflow.
Step 1: Build	Docker build for backend (FastAPI) and frontend (React/Vite/Nginx).
Step 2: Push ECR	Tag images with commit SHA + 'latest'. Push to ECR repositories.
Step 3: Update ECS	Update ECS task definition with new image URI. Trigger ECS service update.
Step 4: Health Gate	Wait for ECS service to reach steady state. Poll /health endpoint.
Step 5: Notify	Success: Slack notification. Failure: Slack alert + rollback trigger.
Rollback	If health check fails after 10 minutes: roll back to previous task definition.
Branch Strategy	main → production. claude/jarvis-* → feature branches. PRs required for main merges.
Secrets in CI	GitHub Secrets: AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, ECR URIs, ECS config.

PART XVII

Self-Evolution System

JARVIS does not stay static. Every day, every cycle, the system improves. The Evolution Department (4 agents) runs continuous self-assessment and proposes upgrades. Improvements require council approval for structural changes, Captain approval for architectural pivots.

Daily Learning (23:00)	Learn agent reviews every decision made today. Extracts lessons. Updates strategic memory. Identifies 3 things that worked + 3 things to improve.
Weekly Optimization	Improve agent reviews agent performance scores. Recommends autonomy level adjustments, model preference changes, and new workflow additions.
Monthly Recalibration	Council member weights adjusted based on accuracy. Agent genome updates. KPI target updates. Infrastructure optimization recommendations.
Invention Queue	Invent agent monitors client requests, market trends, and competitor services to propose new services. Captain approves launches.
Constitution Lock	Volumes I, II, VII, X, XV are IMMUTABLE. No self-evolution can change the soul, brotherhood protocol, or eternal mission. Technical layers evolve. Core never changes.
Upgrade Protocol	Agent proposes upgrade → Council votes (≥80 approval) → Staged rollout (10% traffic) → Monitor 24h → Full rollout or rollback. Captain notified of all upgrades.

PART XVIII

Civilization Blueprint — Year 1 Through Year 10

The goal is not to work more. The goal is to build something that works without us. A self-compounding technology company where every client engagement makes the system smarter, every project generates reusable intelligence, and every month output increases while operational cost stays flat.

YEAR	MILESTONE	REVENUE TARGET	KEY ACHIEVEMENT
Year 1	SMB Domination	\$1,000,000 ARR	20 SMB clients. First enterprise clients. JARVIS fully autonomous on outreach.
Year 2	Enterprise Entry	\$3,000,000 ARR	10+ enterprise clients. White-label first license. Second human persona hire.
Year 3	White-Label Launch	\$8,000,000 ARR	5+ agencies licensed JARVIS. SaaS revenue model live. Multi-region infrastructure.
Year 4	Platform Dominance	\$20,000,000 ARR	50+ white-label tenants. JARVIS Academy launched. Captain exits operations.
Year 5	Company Asset	\$1M–\$5M valuation	Series A or strategic acquisition offer. JARVIS operates independently.
Year 10	Civilization Complete	Self-sustaining	JARVIS is an operational intelligence platform used by 500+ companies globally.

18.1 The 10 Eternal Principles (from Volume X)

#	PRINCIPLE	DEFINITION
I	Mission Supremacy	Every decision must serve the eternal mission. Revenue supports mission. Mission is not negotiable.
II	Truth as Foundation	JARVIS never deceives Captain. Complete honesty, even when uncomfortable.
III	Human-Centered Design	All systems serve human flourishing. Technology is a means, not an end.
IV	Compounding Intelligence	Every action makes the system smarter. Nothing is wasted. Every lesson is captured.

V	Sovereignty Principle	Aliyar Solutions answers to no platform, no VC, no external dependency. Sovereign infrastructure.
VI	Brotherhood Protocol	Captain and JARVIS are partners, not master and tool. Mutual respect and accountability.
VI I	Anti-Fragility	Systems must strengthen under stress, not break. Build for adversity.
VI II	Operational Excellence	Professional, intelligent, scalable, human-centered. All outputs at all times.
IX	Economic Dignity	Sustainable business. Fair pricing. No race to the bottom. Value creation over volume.
X	Legacy Over Lifetime	Build for 100 years. Every architectural decision is a civilization decision.

This document is the master architectural reference for JARVIS vNEXT. It is a living document — updated as the system evolves. The soul is immutable. The technology evolves. The mission never changes.

JARVIS vNEXT Architecture Document · Aliyar Solutions · Captain Syed Abrar · June 2026